



Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Calculus for Engineers

Course Code: MA101L

Course Coordinator Name: Dr. Kuldeep Sharma

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1	Apply the concept of partial differentiation in application of homogeneous and composite functions.	PO1,PO2,PO3,PO8,PO12	Apply	C,P
2	Apply knowledge of partial differentiation in extrema, series expansion of functions and Jacobians.	PO1,PO2,PO3,PO8,PO12	Apply	C,P
3	Apply the concept of vector differentiation in engineering problems.	PO1,PO2,PO3,PO8,PO12	Apply	C,P
4	Employ the concept of multiple integration to find the area of bounded region	PO1,PO2,PO3,PO8,PO12	Apply	C,P
5	Apply the concept of analytic and harmonic functions of complex variables in transformation	PO1,PO2,PO3,PO8,PO12	Apply	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Pinky Saxena			
Dr Sanjay Garg			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Calculus for Engineers

Course Code: MA101L

Course Coordinator Name: Dr. Kuldeep Sharma

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	2	2	-	-	-	-	1	-	-	-	2	-	-
CO2	2	2	2	-	-	-	-	1	-	-	-	2	-	-
CO3	3	2	2	-	-	-	-	1	-	-	-	1	-	-
CO4	2	2	2	-	-	-	-	1	-	-	-	1	-	-
CO5	2	2	2	-	-	-	-	1	-	-	-	1	-	-
PO Target	2.2	2	2	-	-	-	-	1	-	-	-	1.4	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr Pinky Saxena			
Dr Sanjay Garg			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Programming for Problem Solving Course Code: IT101L

Course Coordinator Name: Mr. Dinesh Kumar

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Apply programming constructs of C language to solve real-world problems.	PO1, PO2, PO5, PO8, PO12, PSO1	Apply	C,P
CO2	Use the concepts of looping, branching, and decision-making statements for a given problem.	PO1, PO2, PO4, PO5, PO8, PO12, PSO1	Apply	C,P
CO3	Develop Solutions to problems using modular programming constructs such as functions and recursion	PO1, PO2, PO4, PO5, PO8, PO12, PSO1	Create	C,P,M
CO4	Demonstrate the ability to write C programs using Array, pointers and strings.	PO1, PO2, PO3, PO4, PO5, PO8, PO12, PSO1	Apply	C,P
CO5	Design a solution to problems using the concepts of structure, union, pointers and files handling	PO1, PO2, PO3, PO4, PO5, PO8, PO12, PSO1	Create	C,P,M

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Rishabh Jain		Prof. Arti Sharma	
Prof. Puneet Kumar Goyal			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Programming for Problem Solving Course Code: IT101L

Course Coordinator Name: Mr. Dinesh Kumar

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	2	-	-	2	-	-	1	-	-	-	2	2	-
CO2	3	2	-	2	2	-	-	1	-	-	-	2	2	-
CO3	3	3	-	2	2	-	-	1	-	-	-	2	2	-
CO4	3	3	2	2	2	-	-	1	-	-	-	2	2	-
CO5	3	3	2	2	2	-	-	1	-	-	-	2	2	-
PO Target	3	3	2	2	2	-	-	1	-	-	-	2	2	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Rishabh Jain		Prof. Arti Sharma	
Prof. Puneet Kumar Goyal			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-2026

Year: 1st

Semester: 1st

Course Name: Computer Organization & Logic Design

Course Code: EC201L

Coordinator Name: Dr Shipra Srivastava

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1	Apply the basics of binary arithmetic and codes in digital system design.	PO1, PO2, PO3, PO4, PO5, PO6	Apply	P
2	Design combinational logic circuits using Boolean functions and gate-level minimization	PO1, PO2, PO3, PO4, PO5, PO6	Analyze	P
3	Design sequential logic circuits, including latches, flip-flops, registers, and counters.	PO1, PO2, PO3, PO4, PO5, PO6	Analyze	P
4	Understand computer organization, including bus architecture, processor organization.	PO1, PO2, PO3, PO4, PO5, PO6	Understand	C
5	Understand memory organization, RAM and Read Only Memory (ROM).	PO1, PO2, PO3, PO4, PO5, PO6	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Harshita Prasad		Dr. Abhas Kanungo	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-2026

Year: 1st

Semester: 1st

Course Name: Computer Organization & Logic Design

Course Code: EC201L

Coordinator Name: Dr Shipra Srivastava

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	2	2	1	1	-	-	-	-	-	-	-	-
CO2	3	3	2	2	1	1	-	-	-	-	-	-	-	-
CO3	3	3	2	2	1	1	-	-	-	-	-	-	-	-
CO4	3	3	2	2	2	2	-	-	-	-	-	-	-	-
CO5	3	3	2	2	1	1	-	-	-	-	-	-	-	-
PO Target	3	3	2	2	1.2	1.2	-	-	-	-	-	-	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Harshita Prasad		Dr. Abhas Kanungo	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Design Thinking

Course Code: IT103L

Course Coordinator Name: Dr. Manish Bhardwaj

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1	Understand the basic requirements of a good design.	PO1 , PO2, PO3, PO4, PO6, PO9, PO10, PO12	Understand	C
2	Empathise and ideate the solutions to problems in his environment.	PO1, PO2, PO3, PO4, PO6, PO9, PO10, PO12	Apply	P
3	Prototype and test the developed solutions.	PO1, PO2, PO3, PO4, PO6, PO9, PO10, PO12	Analyze	P
4	Apply the principles of design thinking on developing innovative solutions to the real world problems.	PO1, PO2, PO3, PO4, PO6, PO9, PO10, PO12	Apply	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Raj Kumar		Prof. Arti Sharma	
Prof. Vivek Kumar Sharma		Prof. Sreesh Gaur	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Design Thinking

Course Code: IT103L

Course Coordinator Name: Dr. Manish Bhardwaj

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	1	2	3	2	-	1	-	-	2	2	-	2	1	-
CO2	1	2	3	2	-	1	-	-	2	2	-	2	1	-
CO3	1	2	3	2	-	1	-	-	2	2	-	2	1	-
CO4	1	2	3	2	-	1	-	-	2	2	-	2	1	-
CO5	1	2	3	2	-	1	-	-	2	2	-	2	1	-
PO Target	1	2	3	2	-	1	-	-	2	2	-	2	1	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Raj Kumar		Prof. Arti Sharma	
Prof. Vivek Kumar Sharma		Prof. Sreesh Gaur	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Design & Realization

Course Code: ME101B

Course Coordinator Name: Mr. Vivek Bhardwaj

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1	Understand the concept of Computer-Aided Design (CAD).	PO5,PO10,PO12	Understand	C,P
2	Apply CAD software to create basic 3D models.	PO1,PO3, PO5, PO10, PO12	Apply	C,P
3	Apply CAD and Additive Manufacturing software for 3D printing.	PO1, PO3, PO5, PO9, PO10, PO12	Apply	C,P
4	Understand the fundamentals of Computer-Aided Manufacturing and CNC machining.	PO5, PO9, PO10, PO12	Understand	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prashant Vashishtha			
Ashok Kumar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Design & Realization

Course Code: ME101B

Course Coordinator Name: Mr. Vivek Bhardwaj

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	-	-	-	2	-	-	-	-	2	-	-	-	3
CO2	2	-	2	-	3	-	-	-	-	2	-	-	-	3
CO3	2	-	2	-	3	-	-	-	2	2	-	-	-	3
CO4	-	-	-	-	2	-	-	-	2	2	-	-	-	3
PO Target	2	-	2	-	2.5	-	-	-	2	2	-	-	-	3

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prashant Vashishtha			
Ashok Kumar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science and Engineering

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Environmental Chemistry

Course Code: CH101L

Course Coordinator Name: Dr. Sweta Shukla

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Apply the knowledge of advanced materials for interdisciplinary applications.	PO1, PO2, PO3, PO4, PO6, PO11, PO12, PSO1, PSO2	Apply	C,P
CO2	Employ the concept of electrochemistry for portable energy devices to provide viable solutions for industrial problems.	PO1, PO2, PO3, PO4, PO6, PO11, PO12, PSO1, PSO2	Apply	C,P
CO3	Apply the insight of environment and resources for sustainable development.	PO1, PO2, PO3, PO4, PO6, PO11, PO12, PSO1, PSO2	Apply	C,P
CO4	Determine the environment related issues, their impacts and provide the sustainable solutions.	PO1, PO2, PO3, PO4, PO6, PO11, PO12, PSO1, PSO2	Understand	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Priyanka Rai			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science and Engineering

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Environmental Chemistry

Course Code: CH101L

Course Coordinator Name: Dr. Sweta Shukla

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	2	1	1	-	1	-	-	-	-	1	2	-	-
CO2	2	2	1	2	-	1	-	-	-	-	1	2	-	-
CO3	2	2	1	1	-	2	-	-	-	-	2	2	-	-
CO4	2	2	1	1	-	2	-	-	-	-	2	2	-	-
PO Target	2	2	1	1.25	2	2	-	-	-	-	1.5	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Priyanka Rai			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-2026

Year: 1st

Semester: 1st

Course Name: Computer Organization & Logic Design Lab

Course Code: EC201P

Coordinator Name: Dr Shipra Srivastava

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Apply the basics of binary arithmetic and codes in digital system design.	PO1, PO2, PO3, PO4, PO5, PO6	Apply	P
CO2	Design combinational logic circuits using Boolean functions and gate-level minimization	PO1, PO2, PO3, PO4, PO5, PO6	Analyze	P
CO3	Design sequential logic circuits, including latches, flip-flops, registers, and counters.	PO1, PO2, PO3, PO4, PO5, PO6	Analyze	P
CO4	Understand computer organization, including bus architecture, processor organization.	PO1, PO2, PO3, PO4, PO5, PO6	Understand	C
CO5	Understand memory organization, RAM and Read Only Memory (ROM).	PO1, PO2, PO3, PO4, PO5, PO6	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Harshita Prasad		. Abhas Kanungo	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-2026

Year: 1st

Semester: 1st

Course Name: Computer Organization & Logic Design Lab

Course Code: EC201P

Coordinator Name: Dr Shipra Srivastava

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	-	-	2	-	-	1	-	-	-	2	2	-
CO2	3	3	-	2	2	-	-	1	-	-	-	2	2	-
CO3	3	3	-	2	2	-	-	1	-	-	-	2	2	-
CO4	3	3	2	2	2	-	-	1	-	-	-	2	2	-
CO5	3	3	2	2	2	-	-	1	-	-	-	2	2	-
PO Target	3	3	2	2	2	-	-	1	-	-	-	2	2	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Harshita Prasad		Dr. Abhas Kanungo	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech.

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Programming for Problem Solving Lab

Course Code:IT101P

Course Coordinator Name: Mr. Dinesh Kumar

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Apply programming constructs of C language to solve real-world problems.	CO1, CO2, CO5, CO8, CO12,PSO1	Apply	C,P
CO2	Use the concepts of looping, branching, and decision-making statements for a given problem.	CO1, CO2, CO4, CO5, CO8, CO12,PSO1	Apply	C,P
CO3	Develop Solutions to problems using modular programming constructs such as functions and recursion.	CO1, CO2, CO4, CO5, CO8, CO12,PSO1	Create	C,P,M
CO4	Demonstrate the ability to write C programs using pointers, strings structures and unions.	CO1, CO2, CO3, CO4, CO5, CO8, CO12,PSO1	Apply	C,P
CO5	Design a solution to problems using the concepts of pointers and files handling.	CO1, CO2,CO3, CO4, CO5, CO8, CO12,PSO1	Create	C,P,M

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Rishabh Jain		Prof. Arti Sharma	
Prof. Puneet Kumar Goyal			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech.

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Programming for Problem Solving Lab

Course Code: IT101P

Course Coordinator Name: Mr. Dinesh Kumar

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	-	-	2	-	-	1	-	-	-	2	2	-
CO2	3	3	-	2	2	-	-	1	-	-	-	2	2	-
CO3	3	3	-	2	2	-	-	1	-	-	-	2	2	-
CO4	3	3	2	2	2	-	-	1	-	-	-	2	2	-
CO5	3	3	2	2	2	-	-	1	-	-	-	2	2	-
PO Target	3	3	2	2	2	-	-	1	-	-	-	2	2	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Rishabh Jain		Prof. Arti Sharma	
Prof. Puneet Kumar Goyal			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Web Designing

Course Code: IT102P

Course Coordinator Name: Mr. Mayank Tyagi

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1.	Understand the concept of layout and structure of Hypertext markup language (HTML)	PO1,PO2,PO3,PO4,PO 5,PO8,PO12,PSO1	Apply	C,P
2.	Apply the integration of Cascading style sheets (CSS) in HTML pages.	PO1,PO2,PO3,PO4,PO 5,PO8,PO12,PSO1	Apply	C,P
3.	Apply the JavaScript concept to process and validate the data of a web page on client Machine.	PO1,PO2,PO3,PO4,PO 5,PO8,PO12,PSO1	Create	C,P
4.	Design the website with the application of HTML, CSS and JavaScript.	PO1,PO2,PO3,PO4,PO 5,PO8,PO12,PSO1	Apply	C,P
5.	Demonstrate practical implementation of HTML, CSS, and JavaScript by developing real-time use case projects.	PO1,PO2,PO3,PO4,PO 5,PO8,PO12,PSO1	Apply/Create	C,P,M

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Mr. Kuldeep Kumar Atariya			
Mr. Anmol Jain			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech
Course Name: Web Designing
CO - PO/PSO/APO Matrix

Academic Session: 2025-26
Course Code: IT102P

Year: 1st
Semester: 1st
Course Coordinator Name: Mr. Mayank Tyagi

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	2	2	2	2	-	-	1	-	-	-	3	1	-
CO2	3	2	2	2	2	-	-	1	-	-	-	3	1	-
CO3	3	2	2	2	2	-	-	1	-	-	-	3	1	-
CO4	3	3	3	3	3	-	-	1	-	-	-	3	1	-
CO5	3	2	3	3	3	-	-	1	-	-	-	3	1	-
PO Target	2.8	2.2	2.4	2.4	2.4	-	-	1	-	-	-	3	1	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Mr. Kuldeep Kumar Atariya			
Mr. Anmol Jain			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Basic Proficiency in French

Course Code: HS105P

Course Coordinator Name: Ms. Marry Saini

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Introduce themselves in the respective language and understand the syllables and numbers	PO9	Understand	C
CO2	Apply their learning in basic conversation and understand the social etiquette of professional world	PO9, PO10, PO12	Understand	C
CO3	Understand how language and culture interact in global context and impact intercultural communication	PO9, PO10, PO12	Understand	C
CO4	Utilize the skills of listening, speaking and non verbal communication in the target language	PO12	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Ms. Marry Saini			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Basic Proficiency in French

Course Code: HS105P

Course Coordinator Name: Ms. Marry Saini

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	-	-	-	-	-	-	-	2	2	-	2	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO3	-	-	-	-	-	-	-	-	1	2	-	3	-	-
CO4	-	-	-	-	-	-	-	-	3	3	-	3	-	-
Target	-	-	-	-	-	-	-	-	2	2.5	-	2.5	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Ms. Marry Saini			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Basic Proficiency in German

Course Code: HS104P

Course Coordinator Name: Ms. Anusree Chatterjee

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	To understand the basic parts of speech of the language, put up basic questions and be able to introduce themselves in respective language and understand the syllables and numbers.	PO9, PO10, PO12	Understand	C
CO2	Understand how language and culture interact in global context and impact intercultural communication.	PO9, PO10, PO12	Understand	C
CO3	Apply their learning in basic conversations, sentence building and understand the social etiquette of professional world	PO9, PO10, PO12	Understand	C
CO4	Utilize the skills of listening, speaking and non- verbal communication in the target language.	PO9, PO10, PO12	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Ms. Anusree Chatterjee			
Ms. Akshita Jain			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year:1st

Semester: 1st

Course Name: Basic Proficiency in German

Course Code:HS104P

Course Coordinator Name:Ms. Anusree Chatterjee

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	-	-	-	-	-	-	-	1	3	-	2	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO3	-	-	-	-	-	-	-	-	3	3	-	2	-	-
CO4	-	-	-	-	-	-	-	-	3	3	-	2	-	-
Target	-		-	-	-	-	-	-	2.25	3	-	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Ms. Anusree Chatterjee		Ms. Akshita Jain	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Basic Proficiency in Japanese

Course Code: HS103P

Course Coordinator Name: Mr. Aayush Sharma

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	To Understand the basic parts of speech of the language, put up basic questions and be able to introduce themselves in respective language and understand the Particles.	PO9, PO10, PO12	Understand	C
CO2	Understand how language and culture interact in global context and impact intercultural communication.	PO9, PO10, PO12	Understand	C
CO3	Apply their learning in basic conversations, sentence building and understand the social etiquette of professional world	PO9, PO10, PO12	Understand	C
CO4	Utilize the skills of listening, speaking and non-verbal communication in the target language.	PO9, PO10, PO12	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Mr. Aayush Sharma			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 1st

Semester: 1st

Course Name: Basic Proficiency in Japanese

Course Code: HS103P

Course Coordinator Name: Mr. Aayush Sharma

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	-	-	-	-	-	-	-	1	3	-	2	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO3	-	-	-	-	-	-	-	-	3	3	-	2	-	-
CO4	-	-	-	-	-	-	-	-	3	3	-	2	-	-
Target	-	-	-	-	-	-	-	-	2.25	3	-	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Mr. Aayush Sharma			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session:2025-26

Year: 1st

Semester:1st

Course Name: Basic Proficiency in Spanish

Course Code:HS106P

Course Coordinator Name: Ms. Shivani Rawat

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Introduce themselves in the respective language and understand the syllables and numbers	PO9, PO10, PO12	Understand	C
CO2	Apply their learning in basic conversation and understand the social etiquette of professional world	PO9, PO10, PO12	Understand	C
CO3	Understand how language and culture interact in global context and impact intercultural communication	PO9, PO10, PO12	Understand	C
CO4	Utilize the skills of listening, speaking and non-verbal communication in the target language.	PO9, PO10, PO12	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Shivani Rawat			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session:2025-26

Year: 1st

Semester:1st

Course Name: Basic Proficiency in Spanish

Course Code:HS106P

Course Coordinator Name: Ms. Shivani Rawat

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	-	-	-	-	-	-	-	1	3	-	2	-	-
CO2	-	-	-	-	-	-	-	-	1	3	-	2	-	-
CO3	-	-	-	-	-	-	-	-	3	3	-	2	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	2	-	-
Target	-	-	-	-	-	-	-	-	1.75	3	-	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Aayush Sharma			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Database System

Course Code: IT301L

Course Coordinator Name: Dr. Gaurav Dubey

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO 1	Acquire knowledge of database design methodology for real-life applications.	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO12, PSO1, PSO2	Understand	C
CO 2	Design an information model using the concept of ER diagram.	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO12, PSO1, PSO2	Create	C,P,M
CO 3	Apply the concept of SQL on real-life databases.	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO12, PSO1, PSO2	Apply	C,P
CO 4	Analyze the redundancy problem in the database and reduce it using normalization.	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO12, PSO1, PSO2	Analyze	C,P
CO 5	Identify the broad range of database management issues including data integrity, security, and recovery transactions, as well as enforce entity integrity, referential integrity, key constraints, and domain constraints on the database.	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO12, PSO1, PSO2	Analyze	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Gaurav Dubey		Dr. Atif Jamshed	
Dr. Sunil Kumar		Ms. Shreela Parekh	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Database System

Course Code: IT301L

Course Coordinator Name: Dr. Gaurav Dubey

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	2	1	1	2	1	1	1	1	1	1	2	1	2
CO2	2	2	3	2	3	1	1	1	2	2	2	2	2	2
CO3	3	3	2	1	3	1	1	1	1	1	1	2	1	2
CO4	3	3	2	2	2	1	1	1	1	1	1	2	1	2
CO5	3	3	2	2	3	2	1	2	1	1	1	2	1	2
PO Target	2.8	2.6	2	2.6	2.6	1.2	1	1.2	1.2	1.2	1.2	2	1.2	2

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Gaurav Dubey		Dr. Atif Jamshed	
Dr. Sunil Kumar		Ms. Shreela Parekh	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Operating System

Course Code: CS206L

Course Coordinator Name: Dr. Akash Goel

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Understand the need, evolution and design issues of various categories of operating systems	PO1, PO2, PO3, PO4, PO5, PO6, PO12, PSO1, PSO2	Understand	C
CO2	Apply different CPU scheduling algorithms and deadlock handling methods.	PO1, PO2, PO3, PO4, PO5, PO6, PO12, PSO1, PSO2	Apply	P
CO3	Analyze the principles of concurrency control and process synchronization problems.	PO1, PO2, PO3, PO4, PO5, PO6, PO12, PSO1, PSO2	Analyze	P
CO4	Analyze various memory management techniques for efficient memory allocation.	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO12, PSO1, PSO2	Analyze	P
CO5	Apply the concept of various I/O management, Disk scheduling and file system.	PO1, PO2, PO3, PO4, PO5, PO6, PO10, PO12, PSO1, PSO2	Apply	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Akash Rajak		Mr. Amit Kumar Singh Sanger	
Mr. Kuldeep Kumar Aratiyan		Dr. Akash Goel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Course Name: Operating System

CO - PO/PSO/APO Matrix

Academic Session: 2025-26

Course Code: CS206L

Year: 2nd

Semester: 3rd

Course Coordinator Name: Dr. Akash Goel

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	2	1	1	1	1	-	-	-	-	-	2	1	2
CO2	3	3	3	3	3	3	-	-	-	-	-	2	2	1
CO3	3	3	2	3	3	3	-	-	-	-	-	2	2	2
CO4	3	3	2	3	3	3	-	-	-	1	-	2	2	2
CO5	3	3	2	2	2	2	-	-	-	1	-	2	1	1
PO Target	3	2.8	2	2.4	2.4	2.4	-	-	-	1	-	2	1.6	1.6

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Akash Rajak		Mr. Amit Kumar Singh Sanger	
Mr. Kuldeep Kumar Aratiyan		Dr. Akash Goel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Object Oriented Programming using Java Course Code: CS301L Course Coordinator Name: Dr. Abhishek Goyal

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1	Implement core Java concepts that model real world entities.	PO1,PO2,PO3,PO4,PO5,PO12,PSO1,	Apply	P
2	Develop programs based on new java features.	PO1,PO2,PO3,PO4,PO5,PO12,PSO1,PSO2	Apply	P
3	Apply a collection framework to build modular java programs.	PO1,PO2,PO3,PO4,PO5,PO12,PSO1,PSO2	Apply	P
4	Construct dynamic web pages using JDBC and JSP.	PO1,PO2,PO3,PO4,PO5,PO11,PO12,PSO1,PSO2	Apply	P
5	Implement web and RESTful Web Services with Spring Boot using Spring Framework concepts.	PO1,PO2,PO3,PO4,PO5,PO11,PO12,PSO1,PSO2	Apply	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Abhishek Goyal		Dr. Atif Jamshed	
Mr. Anmol Jain		Dr. Akash Goel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Object Oriented Programming using Java Course Code: CS301L Course Coordinator Name: Dr. Abhishek Goyal

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	3	2	2	2	-	-	-	-	-	-	2	-	-
CO2	2	3	2	2	2	-	-	-	-	-	-	2	2	2
CO3	2	3	2	2	2	-	-	-	-	-	-	3	2	2
CO4	2	2	2	2	3	-	-	-	-	-	2	3	2	2
CO5	2	3	3	3	2	-	-	-	-	-	2	3	2	2
PO Target	2	2.8	2.2	2.2	2.2	-	-	-	-	-	2	2.6	2	2

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Abhishek Goyal		Dr. Atif Jamshed	
Mr. Anmol Jain		Dr. Akash Goel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Probability and Statistics

Course Code: MA105L

Course Coordinator Name: Dr Ruchi Kushwaha

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1	Employ the concept of measure central tendency and regression analysis.	PO1, PO2, PO3, PO8, PO12	Apply	C,P
2	Apply knowledge of probability on distribution function.	PO1, PO2, PO3, PO8, PO12	Apply	C,P
3	Apply the concept of probability density function and normal distribution.	PO1, PO2, PO3, PO8, PO12	Apply	C,P
4	Apply the concept of Markov analysis.	PO1, PO2, PO3, PO8, PO12	Apply	C,P
5	Employ the knowledge of hypothesis by means of Chi-square and ANOVA test.	PO1, PO2, PO3, PO8, PO12	Apply	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr Ruchi Kushwaha			
Dr Vandana			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Probability and Statistics

Course Code: MA105L

Course Coordinator Name: Dr Ruchi Kushwaha

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	2	2	-	-	-	-	1	-	-	-	1	-	-
CO2	2	2	2	-	-	-	-	1	-	-	-	1	-	-
CO3	2	2	1	-	-	-	-	1	-	-	-	1	-	-
CO4	2	2	1	-	-	-	-	1	-	-	-	1	-	-
CO5	2	2	2	-	-	-	-	1	-	-	-	1	-	-
PO Target	2	2	2	-	-	-	-	1	-	-	-	1	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr Ruchi Kushwaha			
Dr Vandana			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Constitution of India

Course Code: HS109L

Course Coordinator Name: Mr. Rajendra Kumar Patel

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Understand basic features and modalities about the Indian constitution.	PO7,PO8,PO12	Understand	F,C
CO2	Clarify the functioning of Indian parliamentary system at the center and state level	PO7,PO8,PO9,PO12	Understand	C
CO3	Understand the aspects of the Indian Legal System and its related bodies.	PO7,PO8,PO9,PO11,PO12	Understand	C
CO4	Apply different laws and regulations related to engineering practices.	PO7,PO8,PO9,PO10, PO11,PO12	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Mr. Rajendra Patel		Mr. Nishant Raj	
Mr. Jatin		Mr. Mohit Singh	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Constitution of India

Course Code: HS109L

Course Coordinator Name: Mr. Rajendra Kumar Patel

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	-	-	-	-	-	1	2				2	-	-
CO2	-	-	-	-	-	-	1	1	1	-	-	2	-	-
CO3	-	-	-	-	-	-	1	1	1	-	1	2	-	-
CO4	-	-	-	-	-	-	1	2	1	1	1	2	-	-
PO Target	c	-	-	-	-	-	1	1.5	1	1	1	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Mr. Rajendra Patel		Mr. Nishant Raj	
Mr. Jatin		Mr. Mohit Singh	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech
Course Name: Soft Skill
Course Outcomes

Academic Session: 2025-26
Course Code: HS111L

Year: 2nd
Semester: 3rd
Course Coordinator Name: Dr. Meetu Kumar

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Demonstrate improved self-awareness and communication skills through structured presentations and vocabulary-building activities.	PO9, PO10, PO12	Understand	C,P
CO2	Apply effective verbal communication techniques, including pronunciation and elevator pitch delivery, to express ideas clearly and confidently	PO9, PO10, PO12	Apply	P
CO3	Exhibit professional behavior, grooming, and teamwork skills in group discussions, interviews, and workplace-related role plays.	PO9, PO10, PO12	Remember	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Meetu Kumar		Mr. Shubham Tyagi	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech
Course Name: Soft Skill
CO - PO/PSO/APO Matrix

Academic Session: 2025-26
Course Code: HS111L

Year: 2nd
Semester: 3rd
Course Coordinator Name: Dr. Meetu Kumar

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO3	-	-	-	-	-	-	-	-	2	3	-	2	-	-
PO Target	-	-	-	-	-	-	-	-	2	3	-	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Meetu Kumar		Mr. Shubham Tyagi	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech
Course Name: Aptitude I
Course Outcomes

Academic Session: 2025-26
Course Code: HS110L

Year: 2nd
Course Coordinator Name: Prof. Shubham Tyagi
Semester: 3rd

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	After completion of the course, the student will be able to, illustrate their comprehension by solving the given problems.	PO1(M), PO2(L), PO4(L), PO6(L), PO12(L)	Understand	C
CO2	After the completion of the course, the student will be able to, apply the learned concepts to new problems and solve them aptly.	PO1(L), PO2(L), PO4(L), PO6(H), PO12(L)	Apply	C
CO3	After the completion of the course, the student will be able to, make use of their thought process to interpret and draw inferences from the given data to reach logical conclusions.	PO1(L), PO2(L), PO4(L), PO6(L), PO12(H)	Analyze	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Shubham Tyagi			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech
Course Name: Aptitude I
CO - PO/PSO/APO Matrix

Academic Session: 2025-26
Course Code:HS110L

Year: 2nd
Semester: 3rd
Course Coordinator Name: Prof. Shubham Tyagi

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	1	-	1	-	1	-	-	-	-	-	1	-	-
CO2	1	1	-	1	-	2	-	-	-	-	-	1	-	-
CO3	1	1	-	1	-	1	-	-	-	-	-	2	-	-
Target	1.33	1	-	1	-	1.33	-	-	-	-	-	1.33	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Shubham Tyagi			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Advance Data Structure

Course Code: CS302B

Course Coordinator Name: Prof. Sreesh Gaur

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Analyze algorithms using asymptotic notations and the Divide & Conquer technique for efficient problem-solving.	PO1, PO2, PO3, PO4, PO8, PO11, PO12, PSO1	Analyze	C,P
CO2	Implement hashing techniques with collision resolution strategies for optimized searching and storage	PO1, PO2, PO3, PO4, PO8, PO11, PO12, PSO1	Apply	C,P
CO3	Construct and manipulate tree-based data structures, including BST, AVL, and Huffman coding	PO1, PO2, PO3, PO4, PO8, PO11, PO12, PSO1	Apply	C,P
CO4	Apply graph representations and traversal techniques like BFS, DFS, Connected Components, and Topological Sorting	PO1, PO2, PO3, PO4, PO8, PO11, PO12, PSO1	Apply	C,P
CO5	Utilize advanced data structures such as Binomial Heaps, priority queue and, Tries for efficient data processing	PO1, PO2, PO3, PO4, PO8, PO11, PO12, PSO1	Apply	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Anurag Mishra		Prof . Pawan Kr Pal	
Prof. Sreesh Gaur		Prof. Rohan Rathore	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Advance Data Structure

Course Code: CS302B

Course Coordinator Name: Prof. Sreesh Gaur

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	3	2	-	-	-	1	-	-	1	2	2	-
CO2	3	3	3	2	-	-	-	1	-	-	1	2	2	-
CO3	3	3	3	2	-	-	-	1	-	-	1	2	2	-
CO4	3	3	3	2	-	-	-	1	-	-	1	2	2	-
CO5	3	3	3	2	-	-	-	1	-	-	1	2	2	-
PO Target	3	3	3	2	-	-	-	1	-	-	1	2	2	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Anurag Mishra		Prof . Pawan Kr Pal	
Prof. Sreesh Gaur		Prof. Rohan Rathore	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Artificial Intelligence and its Applications

Course Code: CS205B

Course Coordinator Name: Dr. Anurag Mishra

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1	Demonstrate foundational understanding of Artificial Intelligence, its key domains, tools, and ethical implications by implementing logic-based systems and classical search algorithms	PO1, PO2, PO3, PO4, PO5, PO6	Apply	C,P
2	Design and simulate intelligent agents and multi-agent systems in diverse environments, incorporating concepts of decision-making, utility, and communication strategies.	PO1, PO2, PO3, PO4, PO5, PO6	Apply	C,P
3	Apply reinforcement learning techniques to develop learning agents and multi-agent environments, and visualize their behavior in real-world-inspired simulations and games.	PO1, PO2, PO3, PO4, PO5, PO6	Apply	C,P
4	Construct and optimize Genetic Algorithm-based solutions to solve complex real-world problems such as scheduling, path planning, and feature selection, while exploring hybrid and advanced evolutionary strategies.	PO1, PO2, PO3, PO4, PO5, PO6	Apply	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Anurag Mishra		Prof. Bhagwan Gupta	
Dr. Gaurav Dubey		Prof. Dheeraj Pandey	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Artificial Intelligence and its Applications

Course Code: CS205B

Course Coordinator Name: Dr. Anurag Mishra

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	2	3	3	1	-	-	-	-	-	2	2	-
CO2	2	3	3	2	3	1	-	-	-	-	-	2	2	-
CO3	2	3	3	2	2	1	-	-	-	-	-	3	2	-
CO4	1	2	2	3	2	3	-	-	-	-	-	3	2	-
PO Target	2	2.75	2.5	2.5	2.5	1.5	-	-	-	-	-	2.5	2	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Anurag Mishra		Prof. Bhagwan Gupta	
Dr. Gaurav Dubey		Prof. Dheeraj Pandey	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech CS

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Database System Lab

Course Code: IT301P

Course Coordinator Name: Dr. Gaurav Dubey

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO 1	Design Logical and Conceptual database schema for real life problem using ERD	PO1, PO2, PO3, PO4, PO8, PO11, PO12, PSO1	Create	C
CO 2	Apply SQL to store, retrieve and manipulate data in relational database.	PO1, PO2, PO3, PO4, PO8, PO11, PO12, PSO1	Apply	C,P,M
CO 3	Apply PL/SQL to solve real world database management and automation task	PO1, PO2, PO3, PO4, PO8, PO11, PO12, PSO1	Apply	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Gaurav Dubey		Dr. Atif Jamshed	
Dr. Sunil Kumar		Ms. Shreela Pareek	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech CS

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Database System Lab

Course Code: IT301P

Course Coordinator Name: Dr. Gaurav Dubey

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	2	3	1	3	1	1	1	2	2	1	2	1	2
CO2	3	3	2	1	3	1	1	1	1	1	1	2	1	2
CO3	3	3	2	1	3	1	1	1	1	2	1	2	1	2
PO Target	2.6	2.6	2.3	1	3	1	1	1	1.3	1.6	1	2	1	2

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Gaurav Dubey		Dr. Atif Jamshed	
Dr. Sunil Kumar		Ms. Shreela Pareek	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Operating System Lab

Course Code: CS206P

Course Coordinator Name: Dr. Akash Goel

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Apply knowledge of basic UNIX System calls and Shell programming.	PO1, PO5, PO8, PO9, PO10, PO1, PO12, PSO1, PSO2	Apply	P
CO2	Implement various CPU scheduling algorithms and deadlock handling techniques.	PO1, PO2, PO3, PO5, PO8, PO9, PO10, PO1, PO12, PSO1, PSO2	Apply	P
CO3	Implement memory management, process synchronization techniques, page replacement techniques, disk scheduling	PO1, PO2, PO5, PO8, PO9, PO10, PO1, PO12, PSO1, PSO2	Apply	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Akash Rajak		Mr. Amit Kumar Singh Sanger	
Mr. Kuldeep Kumar Aratiyan		Dr. Akash Goel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Operating System Lab

Course Code: CS206P

Course Coordinator Name: Dr. Akash Goel

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	-	-	-	3	-	-	1	1	1	1	2	2	1
CO2	3	3	1	2	2	-	-	1	2	2	1	2	2	1
CO3	3	3	-	2	2	-	-	1	2	2	1	2	2	2
PO Target	3	3	1	2	2.33	-	-	1	1.67	1.67	1	2	2	1.33

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Akash Rajak		Mr. Amit Kumar Singh Sanger	
Mr. Kuldeep Kumar Aratiyan		Dr. Akash Goel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Object Oriented Programming using Java Lab

Course Code: CS301P

Course Coordinator Name: Dr. Abhishek Goyal

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
1	Perform Java oops concepts on an integrated development environment to solve real world problems.	PO1,PO2,PO3,PO4,PO 5,PO12,PSO1,PSO2	Apply	P
2	Solve problems in context of programming code based on collections and new java features.	PO1,PO2,PO3,PO4,PO 5,PO12,PSO1,PSO2	Apply	P
3	Develop a solution for case study-based problem using advance java concepts.	PO1,PO2,PO3,PO4,PO 5,PO11,PO12,PSO1,P SO2	Apply	P
4	Design RESTful Web Services with Spring Boot Test using Spring Framework concepts.	PO1,PO2,PO3,PO4,PO 5,PO11,PO12,PSO1,P SO2	Apply	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Abhishek Goyal		Dr. Atif jamshed	
Mr. Anmol Jain		Dr. Akash Goel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Object Oriented Programming using Java Lab

Course Code: CS301P

Course Coordinator Name: Dr. Abhishek Goyal

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	3	2	2	2	-	-	-	-	-	-	2	2	2
CO2	2	3	2	2	2	-	-	-	-	-	-	3	2	2
CO3	2	2	2	2	3	-	-	-	-	-	2	3	2	2
CO4	2	3	3	3	2	-	-	-	-	-	2	3	2	2
PO Target	2	2.75	2.25	2.25	2.25	-	-	-	-	-	2	2.75	2	2

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Abhishek Goyal		Dr. Atif jamshed	
Mr. Anmol Jain		Dr. Akash Goel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Social Internship Assessment

Course Code: IT105P

Course Coordinator Name: Prof. Mohit Singh Tanwar

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Demonstrate an understanding of social issues and challenges by applying academic knowledge to real-world community service activities.	PO1, PO2, PO3, PO4, PO6, PO7, PO8, PO9, PO10, PO12	Create	C
CO2	Develop leadership, communication, and teamwork skills through active participation, coordination, and interaction with community organizations.	PO1, PO2, PO3, PO6, PO7, PO8, PO9, PO10, PO11, PO12	Create	C
CO3	Critically analyze the impact of social interventions and reflect on personal and professional growth through structured reporting and presentation.	PO1, PO2, PO3, PO4, PO6, PO7, PO8, PO9, PO10, PO11, PO12	Analyze	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof.Mohit Singh Tanwar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 2nd

Semester: 3rd

Course Name: Social Internship Assessment

Course Code: IT105P

Course Coordinator Name: Prof. Mohit Singh Tanwar

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	2	1	1	-	3	3	2	2	2	-	2	-	-
CO2	1	1	2	-	-	2	2	3	3	3	2	2	-	-
CO3	2	2	1	1	-	3	3	3	3	3	1	2	-	-
PO Target	1.67	1.67	1.33	1	-	2.67	2.67	2.67	2.67	2.67	1.5	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof.Mohit Singh Tanwar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Database Management System

Course Code: BCS501

Course Coordinator Name: Dr Sunil Kr

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Apply database knowledge to design solutions for real life problems.	PO,PO2,PO3, PO4,PO5,PO7,PO10,PO11,PO12,PSO1	Apply	C,P
CO2	Apply query processing techniques using SQL and PL/SQL to automate the real time problems of databases.	PO1, PO2, PO3,PO4,PO5,PO10,PO11,PO12,PSO1	Apply	C,P
CO3	Solve the redundancy problem in database tables using normalization.	PO1, PO2, PO3,PO4,PO5,PO9,PO10,PO11,PO12, PSO1	Understand	C
CO4	Understand the concepts of transactions and recovery schemes.	PO1, PO2, PO3, PO4,PO5,PO8,PO11,PO12,PSO1	Apply	C,P
CO5	Understand the concepts of concurrency control techniques.	PO1, PO2, PO3, PO4,PO5,PO6,PO8,PO11,PO12,PSO1	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr Ajay kumar Shrivastava		Prof. Shivani	
Dr. Sunil Kumar		Prof. Puneet Kr Goyal	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Database Management System

Course Code: BCS501

Course Coordinator Name: Dr Sunil Kr

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	2	3	3	2	-	1	-	-	2	1	2	1	-
CO2	3	2	3	2	3	-	-	-	-	2	2	2	2	-
CO3	3	2	3	2	2	-	-	-	2	2	2	2	2	-
CO4	3	3	3	2	3	-	-	1	-	-	1	2	2	-
CO5	3	3	3	2	3	1	-	1	-	-	1	2	2	-
PO Target	3	2.4	3	2.2	2.6	1	1	1	2	2	1.4	2	1.8	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr Ajay kumar Shrivastava		Prof. Shivani	
Dr. Sunil Kumar		Prof. Puneet Kr Goyal	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Web Technology

Course Code: BCS502

Course Coordinator Name: Ms. Shivani

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Understand the fundamentals of web development with HTML and XML.	PO1,PO2,PO3,PO4,PO5,PO8,PO9,PO10,PO11,PO12,PSO1	Understand	C
CO2	Apply CSS to design responsive web applications.	PO1,PO2,PO3,PO4,PO5,PO8,PO9,PO10,PO11,PO12,PSO1	Apply	P
CO3	Apply JavaScript, AJAX for scripting HTML documents and networking concepts required for a website.	PO1,PO2,PO3,PO4,PO5,PO8,PO9,PO10,PO11,PO12,PSO1	Apply	P
CO4	Implement server-side applications using EJB & Node.js with MongoDB.	PO1,PO2,PO3,PO4,PO5,PO8,PO9,PO10,PO11,PO12,PSO1	Apply	P
CO5	Apply components of Servlets and Java Server Pages(JSP) to handle HTTP requests and session tracking.	PO1,PO2,PO3,PO4,PO5,PO8,PO9,PO10,PO11,PO12,PSO1	Apply	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Abhishek Goyal		Mr. Amit Kumar Sanger	
Ms. Shivani		Mr. Nikhil Saraswat	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Web Technology

Course Code: BCS502

Course Coordinator Name: Ms. Shivani

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	2	2	1	2	-	-	1	2	1	2	3	1	-
CO2	3	2	3	1	3	-	-	1	2	1	2	3	1	-
CO3	3	3	3	2	3	-	-	1	2	1	2	3	2	-
CO4	3	3	3	2	3	-	-	1	2	1	2	3	2	-
CO5	3	3	3	3	3	-	-	1	2	1	2	3	2	-
PO Target	c	2.6	2.8	1.8	2.8	-	-	1	2	1	2	3	1.6	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Abhishek Goyal		Mr. Amit Kumar Sanger	
Ms. Shivani		Mr. Nikhil Saraswat	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Design and Analysis of Algorithm

Course Code: BCS503

Course Coordinator Name: Prof. Vivek Sharma

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO-1	Analyze the performance of algorithms using different asymptotic analysis methods.	PO1,PO2,PO3,PO4,PO5,PO12, PSO1	Analyze	C,P
CO-2	Apply advanced data structures to solve computational problems.	PO1,PO2,PO3,PO4,PO5,PO11,PO12, PSO1	Apply	C,P
CO-3	Apply the various algorithmic paradigms, including divide & conquer and greedy algorithms to solve the problems.	PO1,PO2,PO3,PO4,PO5,PO11,PO12, PSO1	Apply	C,P
CO-4	Analyze the solutions for optimization problems using dynamic programming and branch & bound techniques.	PO1,PO2,PO3,PO4,PO5,PO11,PO12, PSO1	Analyze	C,P
CO-5	Understand the concepts of NP-completeness, Randomized and Approximation Algorithms.	PO1,PO2,PO3,PO4,PO5,PO11,PO12PSO1	Understand	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Raj Kumar		Prof. Vivek Kumar Sharma	
Prof. Vikas Gangwar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Design and Analysis of Algorithm

Course Code: BCS503

Course Coordinator Name: Prof. Vivek Sharma

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	3	1	1	-	-	-	-	-	1	1	3	-
CO2	2	3	3	2	1	-	-	-	-	-	1	1	3	-
CO3	2	3	2	2	1	-	-	-	-	-	2	2	3	-
CO4	3	2	3	3	2	-	-	-	-	-	2	2	3	-
CO5	2	2	2	1	1	-	-	-	-	-	1	1	3	-
PO Target	2.4	2.6	2.6	1.8	1.2	-	-	-	-	-	1.4	1.4	3	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Raj Kumar		Prof. Vivek Kumar Sharma	
Prof. Vikas Gangwar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-2026

Year: 3rd

Semester: 5th

Course Name: Data Analytics

Course Code: BCS052

Course Coordinator Name: Prof. Mohit Singh Tanwar

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Discuss the life cycle phases of Data Analytics through discovery, planning and building.	PO1,PO12,PSO1	Understand	C
CO2	Apply various Data Analysis Techniques.	PO1,PO4,PO5,PO10,PO12, PSO12	Apply	P
CO3	Apply mining techniques on streaming data.	PO1,PO4,PO5,PO10,PO12, PSO12	Apply	P
CO4	Compare different clustering and frequent pattern mining algorithms.	PO1,PO4,PO2,PO5,PO10,PO12,PSO12	Analyze	P
CO5	Apply R tool for developing and evaluating real time applications.	PO1,PO4,PO5,PO10,PO12, PSO12	Apply	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Mohit Singh Tanwar		Prof. Tarul	
Prof. Dushyant		Prof. Rahul	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-2026

Year: 3rd

Semester: 5th

Course Name: Data Analytics

Course Code: BCS052

Course Coordinator Name: Prof. Mohit Singh Tanwar

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	-	-	-	-	-	-	-	-	-	-	2	1	-
CO2	2	-	-	2	1	-	-	-	-	1	-	2	2	-
CO3	2	-	-	2	1	-	-	-	-	1	-	2	2	-
CO4	2	2	-	2	1	-	-	-	-	1	-	2	2	-
CO5	2	-	-	2	1	-	-	-	-	2	-	2	2	-
PO Target	2	2	-	2	1	-	-	-	-	1.25	-	2	1.8	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Mohit Singh Tanwar		Prof. Tarul	
Prof. Dushyant		Prof. Rahul	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name:

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Machine Learning Techniques Course Code: BCS055

Course Coordinator Name: Prof. Bhagvan Krishna Gupta

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Understand the need for machine learning for problem-solving.	PO1, PO2, PO3, PO4, PO5, PO12, PSO1	Understand	C
CO2	Apply machine learning techniques for solving different real-world problems.	PO1, PO2, PO3, PO4, PO5, PO12, PSO1	Apply	C,P
CO3	Apply Decision tree and instance-based learning in solving complex problems.	PO1, PO2, PO3, PO4, PO5, PO12, PSO1	Apply	C,P
CO4	Apply ANN and DL to complex engineering problems.	PO1, PO2, PO3, PO4, PO5, PO12, PSO1	Apply	C,P
CO5	Apply reinforcement learning and genetic algorithms to real-world applications.	PO1, PO2, PO3, PO4, PO5, PO12, PSO1	Apply	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Bhagvan Krishna Gupta		Prof. Rajendra Patel	
Prof. Nishant Raj		Prof. Kirti Sharma	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name:

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Machine Learning Techniques Course Code: BCS055

Course Coordinator Name: Prof. Bhagvan Krishna Gupta

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	2	2	2	-	-	-	-	-	-	1	2	-
CO2	3	3	3	3	2	-	-	-	-	-	-	1	2	-
CO3	3	3	3	3	2	-	-	-	-	-	-	1	2	-
CO4	3	3	3	3	2	-	-	-	-	-	-	1	2	-
CO5	3	3	3	3	2	-	-	-	-	-	-	1	2	-
PO Target	3	3	2.8	2.8	2	-	-	-	-	-	-	1	2	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Bhagvan Krishna Gupta		Prof. Rajendra Patel	
Prof. Nishant Raj		Prof. Kirti Sharma	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name:

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Constitution of India

Course Code: BNC501

Course Coordinator Name: Prof. Swati

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Identify and explore the basic features and modalities about Indian constitution.	PO7,PO8,PO12	Apply	F,C
CO2	Differentiate and relate the functioning of Indian parliamentary system at the center and state level.	PO7,PO8,PO9,PO12	Apply	F,C
CO3	Differentiate different aspects of Indian Legal System and its related bodies	PO7,PO8,PO9,PO11,PO12	Apply	F,C
CO4	Discover and apply different laws and regulations related to engineering practices.	PO7,PO8,PO9,PO10,PO11, PO12	Analyze	F,C
CO5	Correlate role of engineers with different organizations and governance models	PO7,PO8,PO9,PO10,PO12	Apply	F,C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Swati		Prof. Kirti Sharma	
Prof. Aditi Joshi		Ms. Tarul	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name:

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Constitution of India

Course Code: BNC501

Course Coordinator Name: Prof. Swati

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	-	-	-	-	-	1	2	-	-	-	2	-	-
CO2	-	-	-	-	-	-	1	1	1	-	-	2	-	-
CO3	-	-	-	-	-	-	1	1	1	-	1	2	-	-
CO4	-	-	-	-	-	-	1	2	1	1	1	2	-	-
CO5	-	-	-	-	-	-	1	2	1	1	-	2	-	-
PO Target	-	-	-	-	-	-	1	1.6	1	1	1	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Swati		Prof. Kirti Sharma	
Prof. Aditi Joshi		Ms. Tarul	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Database Management System Lab

Course Code: BCS551

Course Coordinator Name: Dr Sunil Kr

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Design logical and conceptual database schema for a real life problem using ERD tool.	PO1,PO2,PO3,PO4,PO5,PO9,PO10,PO11,PO12,PSO1	Create	C, P, M
CO2	Implement queries in SQL to store, retrieve, and manipulate data in relational databases.	PO1,PO2,PO3,PO5,PO9,PO10,PO11,PO12,PSO1	Create	C, P, M
CO3	Apply PL/SQL to solve real-world database management and automation tasks.	PO1,PO2,PO3,PO4,PO5,PO9,PO10,PO11,PO12,PSO1	Apply	C, P, M

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Mr. Puneet kr Goyal		Dr.Sunil Kumar	
Ms. Shivani			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Database Management System Lab

Course Code: BCS551

Course Coordinator Name: Dr Sunil kr

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	3	2	3	-	-	-	2	2	1	2	3	-
CO2	3	2	2	-	3	-	-	-	2	2	1	2	3	-
CO3	3	3	3	2	3	-	-	-	2	2	1	2	3	-
PO Target	3	2.66	2.66	2	3	-	-	-	2	2	1	2	3	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Mr. Puneet kr Goyal		Dr. Sunil Kumar	
Ms. Shivani			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Web Technology Lab

Course Code: BCS552

Course Coordinator Name: Prof. Shivani

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Develop dynamic and responsive website by using HTML, CSS, JavaScript and XML.	PO1,PO2,PO3,PO4,PO5, PO8,PO9,PO10,PO11,P O12,PSO1	Apply	P
CO2	Develop web application with MongoDB by using different components of Java Bean and Node.js	PO1,PO2,PO3,PO4,PO5, PO8,PO9,PO10,PO11,P O12,PSO1	Apply	P
CO3	Construct server-side java application using Servlet & JSP tools to process request and response data.	PO1,PO2,PO3,PO4,PO5, PO8,PO9,PO10,PO11,P O12,PSO1	Apply	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Abhishek Goyal		Prof. Amit Kumar Singh Sanger	
Prof. Shivani		Prof. Nikhil Saraswat	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Web Technology Lab

Course Code: BCS552

Course Coordinator Name: Prof. Shivani

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	3	3	3	-	-	1	2	1	2	3	3	-
CO2	3	3	3	3	3	-	-	1	2	1	2	3	3	-
CO3	3	3	3	3	3	-	-	1	2	1	2	3	3	-
PO Target	3	3	3	3	3	-	-	1	2	1	2	3	3	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Abhishek Goyal		Prof. Amit Kumar Singh Sanger	
Prof. Shivani		Prof. Nikhil Saraswat	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean



Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Design and Analysis of Algorithm Lab

Course Code: BCS553

Course Coordinator Name: Prof. Vivek Sharma

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Illustrate basic techniques for designing algorithms, including the techniques of recursion and iterative approach	PO1,PO2,PO3,PO4,PO5,PO11,PO12, PSO1	Apply	C,P
CO2	Apply algorithms to solve problems using various algorithm design strategies.	PO1,PO2,PO3,PO4,PO5,PO11,PO12, PSO1	Apply	C,P
CO3	Analyze the performance of algorithms with respect to time and space complexity	PO1,PO2,PO3,PO4,PO5,PO11,PO12, PSO1	Analyze	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Raj Kumar		Prof. Vivek Kumar Sharma	
Prof. Vikas Gangwar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Design and Analysis of Algorithm Lab

Course Code: BCS553

Course Coordinator Name: Prof. Vivek Sharma

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	1	2	2	1	1	-	-	-	-	-	1	2	3	-
CO2	1	3	2	1	2	-	-	-	-	-	1	2	3	-
CO3	2	3	3	2	3	-	-	-	-	-	2	2	3	-
PO Target	1.33	2.67	2.33	1.33	2.00	-	-	-	-	-	1.33	2.00	3.00	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Dr. Raj Kumar		Prof. Vivek Kumar Sharma	
Prof. Vikas Gangwar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Mini Project or Internship Assessment Course Code:BCS554 Course Coordinator Name: Prof. Rajendra Patel

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Identify a problem and gather its requirements	PO1, PO2, PO3, PO4, PO5, PO6, PO9, PO10, PO11, PO12, PSO1, PSO2	Analyze	C
CO2	Design a solution of the problem using latest tools & techniques.	PO1, PO2, PO3, PO4, PO5, PO9, PO11, PO12, PSO1, PSO2	Understand	C
CO3	Develop a project using latest technology.	PO1, PO2, PO3, PO4, PO5, PO6, PO9, PO11, PO12, PSO1, PSO2	Apply	C
CO4	Develop professional skills and critical thinking to prepare for major project	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2	Apply	C
CO5	Demonstrate an ability to present project works to the evaluators.	PO1, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Rajendra Patel			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 5th

Course Name: Mini Project or Internship Assessment

Course Code: BCS554

Course Coordinator Name: Prof. Rajendra Patel

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	3	3	3	2	-	-	2	2	2	3	2	2
CO2	3	3	3	3	3	-	-	-	2	-	2	3	2	3
CO3	3	3	3	3	3	1	-	-	3	-	2	2	3	2
CO4	3	3	3	3	3	2	-	2	3	2	2	3	2	2
CO5	2	-	-	-	-	-	-	2	2	3	2	3	2	2
PO Target	2.8	3	3	3	3	1.67	-	2	2.4	2.33	2	2.80	2.2	2.2

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Rajendra Patel			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: Btech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Artificial Intelligence

Course Code:BCS701

Course Coordinator Name: Ms. Shreela Pareek

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Understand the concepts of artificial intelligence and intelligent agents.	PO1, PO6, PO12, PSO1, PSO2	Understand	C
CO2	Apply principles of AI in problem-solving.	PO1, PO2, PO3, PO4, PO5, PO12, PSO1, PSO2	Apply	C,P
CO3	Determine the effectiveness of truths by knowledge representation methods in AI.	PO1, PO2, PO3, PO4, PO5, PO12, PSO1, PSO2	Analyze	C,P
CO4	Analyze uncertainty in knowledge representation using probabilistic reasoning, fuzzy logic, and basic neural network concepts.	PO1, PO2, PO3, PO4, PO5, PO8, PO11, PO12, PSO1	Analyze	C,P
CO5	Analyze various AI applications in Information retrieval and extraction, Natural Language Processing, speech recognition and Robots.	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO12, PSO1	Analyze	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Ms. Shreela Pareek		Mr. Siddharth Shravan Jha	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: Btech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Artificial Intelligence

Course Code:BCS701

Course Coordinator Name: Ms. Shreela Pareek

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	-	-	-	-	2	-	-	-	-	-	3	1	1
CO2	3	3	2	2	3	-	-	-	-	-	-	3	2	1
CO3	3	3	2	3	3	-	-	-	-	-	-	2	2	1
CO4	3	3	3	3	3	-	-	2	-	-	2	2	1	-
CO5	3	3	2	3	3	3	-	2	-	-	-	2	2	-
PO Target	3	3	2.25	2.75	3	2.5	-	2	-	-	2	2.4	1.6	1

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Ms. Shreela Pareek		Mr. Siddharth Shravan Jha	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Cloud Computing

Course Code: BCS071

Course Coordinator Name: Prof. Anmol Jain

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Understand the evolution & principles of cloud computing	PO1, PO2, PO5, PO9, PO12, PSO2	Understand	C
CO2	Apply Virtualization of hardware and software resources for Cloud Computing.	PO1, PO2, PO5, PO8, PO9, PO10, PO12, PSO1, PSO2	Apply	P
CO3	Implement data access management, data storage and computing services on Cloud.	PO1, PO2, PO5, PO8, PO9, PO10, PO12, PSO1, PSO2	Apply	P
CO4	Explain Inter cloud resources management, cloud storage services and Security Services.	PO1, PO2, PO5, PO8, PO9, PO10, PO12, PSO2	Analyze	P
CO5	Analyze standards, and applications of advanced cloud technologies.	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2	Analyze	P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Anmol Jain		Prof. Rahul	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Cloud Computing

Course Code: BCS071

Course Coordinator Name: Prof. Anmol Jain

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	1	-	-	2	-	-	-	1	-	-	2	2	-
CO2	2	1	-	-	2	-	-	1	1	1	-	2	2	-
CO3	2	1	-	-	2	-	-	1	1	1	-	2	2	-
CO4	3	2	-	-	2	-	-	1	1	1	-	2	2	-
CO5	3	3	2	2	3	-	-	2	2	1	2	2	3	-
PO Target	2.4	1.6	2	2	2.2	-	-	1.25	1.2	1	2	2	2.2	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Anmol Jain		Prof. Rahul	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name:

Academic Session:2025-26

Year: 4th

Semester:7th

Course Name: Renewable Energy Resources

Course Code: BOE074

Course Coordinator Name: Prof. Swati

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Understand the significance of various non-conventional energy resources, their availability and limitations, working of solar cell, its material, advantages and limitations	PO1, PO2, PO3, PO7, PO9, PO12	Understand	F,C
CO2	Apply the knowledge to select suitable solar thermal collectors to meet desired need within realistic constraints such as economic, environmental, and sustainability	PO1, PO2, PO3, PO7, PO9, PO12	Apply	F,C,P
CO3	Understand the system and working of non conventional energy resources such as Magneto-hydrodynamics (MHD) generator, geothermal and fuel cell	PO1, PO2, PO3, PO7, PO9, PO12	Understand	F,C,P
CO4	Analyze the optimum power generation through wind power plant and understand the system and working of thermo-electric and thermo-ionic systems	PO1, PO2, PO3, PO7, PO9, PO12	Analyze	F,C,P
CO5	Understand the basic systems of Ocean thermal energy conversion, wave energy plant, biomass energy system to meet the energy shortage requirement	PO1, PO2, PO3, PO7, PO9, PO12	Understand	F,C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Swati		Prof. Dushyant Kumar	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name:

Academic Session:2025-26

Year: 4th

Semester:7th

Course Name: Renewable Energy Resources

Course Code: BOE074

Course Coordinator Name: Prof. Swati

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	1	1	1	-	-	-	2	-	1	-	-	2	-	-
CO2	2	1	2	-	-	-	2	-	1	-	-	2	-	-
CO3	1	1	1	-	-	-	2	-	1	-	-	2	-	-
CO4	3	2	2	-	-	-	2	-	1	-	-	2	-	-
CO5	1	1	1	-	-	-	2	-	1	-	-	2	-	-
PO Target	1.6	1.2	1.4	-	-	-	2	-	1	-	-	2	-	-

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Swati		Prof. Dushyant Kumar	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Artificial Intelligence Lab

Course Code:BCS751

Course Coordinator Name: Ms. Shreela Pareek

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Identify problems of artificial intelligence techniques.	PO1, PO3, PO4,PO5,PO6, PO12, PSO1,PSO2	Apply	F,C,P
CO2	Judge applicability of AI techniques in real-world problems.	PO1,PO2, PO3, PO4,PO5,PO6, PO12, PSO1,PSO2	Apply	C,P
CO3	Design the basic intelligent systems.	PO1,PO2,PO3,PO5,PO6, PO12, PSO1,PSO2	Apply	C,P,M

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Ms. Shreela Pareek		Ms. Swati	
Mr. Siddharth Shravan Jha			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Artificial Intelligence Lab

Course Code:BCS751

Course Coordinator Name: Ms. Shreela Pareek

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	-	3	2	2	2	-	-	-	-	-	1	2	1
CO2	2	2	3	3	2	3	-	-	-	-	-	1	3	1
CO3	3	2	2	-	2	3	-	-	-	-	-	1	3	1
PO Target	2.3	2.0	2.67	2.5	2	2.67	-	-	-	-	-	1	2.67	1

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Ms. Shreela Pareek		Ms. Swati	
Mr. Siddharth Shravan Jha			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 7th

Course Name: Mini Project and Internship Assessment

Course Code: BCS752

Course Coordinator Name: Prof. Aditi

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Identify a problem and gather its requirements	PO1, PO2, PO3, PO4, PO5, PO6, PO9, PO10, PO11, PO12, PSO1, PSO2	Analyze	C
CO2	Design a solution of the problem using latest tools & techniques	PO1, PO2, PO3, PO4, PO5, PO9, PO11, PO12, PSO1, PSO2	Understand	C
CO3	Develop a project using latest technology.	PO1, PO2, PO3, PO4, PO5, PO6, PO9, PO11, PO12, PSO1, PSO2	Apply	C
CO4	Develop professional skills and critical thinking to prepare for major project	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2	Apply	C
CO5	Demonstrate an ability to present project works to the evaluators.	PO1, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2	Understand	C

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Aditi		Prof. Nikhil Saraswat	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 3rd

Semester: 7th

Course Name: Mini Project and Internship Assessment

Course Code: BCS752

Course Coordinator Name: Prof. Aditi

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	3	3	3	3	2	-		2	2	2	3	2	2
CO2	3	3	3	3	3	-	-		2	-	2	3	2	3
CO3	3	3	3	3	3	1	-		3	-	2	2	3	2
CO4	3	3	3	3	3	2	-	2	3	2	2	3	2	2
CO5	2	-	-	-	-	-	-	2	2	3	2	3	2	2
PO Target	2.8	3	3	3	3	1.67	-	2	2.4	2.33	2	2.80	2.2	2.2

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Aditi		Prof. Nikhil Saraswat	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B Tech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Project I

Course Code: BCS753

Course Coordinator Name: Prof. Sreesh Gaur

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Select and summarize all aspects of real-life problems through information gathering.	PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2	Understand	C,P
CO2	Apply acquired knowledge to develop a Conceptual model.	PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2	Apply	C,P
CO3	Analyze the outcome of each phase using various tools and techniques.	PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2	Analyze	C,P
CO4	Defend the validity of idea or quality of result with the previous data/ result.	PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2	Evaluate	C,P
CO5	Test the working model and demonstrate the results by publishing the idea/outcome.	PO2, PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12, PSO1, PSO2	Create	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Sreesh Gaur		Prof. Rahul	
Prof. Mohit S Tanwar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B Tech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Project I

Course Code: BCS753

Course Coordinator Name: Prof. Sreesh Gaur

:

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	-	3	3	3	3	1	2	-	3	2	3	3	3	3
CO2	-	3	3	3	2	1	2	-	3	2	3	3	3	3
CO3	-	3	3	3	2	1	2	-	3	2	3	3	3	3
CO4	-	3	3	3	2	1	2	-	3	2	2	2	3	3
CO5	-	3	3	3	2	1	2	-	3	2	1	2	3	3
PO Target	-	3	3	3	2.2	1	2	-	3	2	2.4	2.6	3	3

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Sreesh Gaur		Prof. Rahul	
Prof. Mohit S Tanwar			

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Start up and Entrepreneurial Activity Assessment Course Code: BCS754 Course Coordinator Name: Prof. Swati

Course Outcomes

After completion of the course, the student will be able to		Relevant POs/ PSOs	Revised Bloom's Level (BL)	Knowledge Category (KC)
CO No.	Statement of Course Outcome			
CO1	Understand the fundamental concepts of entrepreneurship, startup ecosystem, and innovation process.	PO1,PO2,PO3,PO6, PO8,PO10,PO11,PO12, PSO2	Understand	C
CO2	Apply entrepreneurial skills to identify, analyze, and evaluate potential business opportunities and prepare a basic business plan.	PO1,PO2,PO3,PO4, PO5,PO9,PO10,PO11, PO12,PSO1,PSO2	Apply	C,P
CO3	Demonstrate the ability to design and present a startup model integrating financial, ethical, and technological aspects.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10,PO11, PO12, PSO1,PSO2	Apply	C,P

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Swati		Prof. Rajendra Patel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean

Department of Computer Science

Program Name: B.Tech

Academic Session: 2025-26

Year: 4th

Semester: 7th

Course Name: Start up and Entrepreneurial Activity Assessment Course Code: BCS754 Course Coordinator Name: Prof. Swati

CO - PO/PSO/APO Matrix

CO No.	Programme Outcome (PO)												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	2	1	1	-	-	1	-	1	-	2	1	1	-	1
CO2	2	2	2	1	1	-	-	-	1	2	2	1	1	2
CO3	1	2	3	1	2	1	1	3	2	3	3	2	2	3
PO Target	1.66	1.66	2	1	1.5	2	1	2	1.5	2.33	2	1.33	1.5	2

Faculty Members Teaching the Course	Signature	Faculty Members Teaching the Course	Signature
Prof. Swati		Prof. Rajendra Patel	

Signature of Course Coordinator

Assoc./ Asst. Head DOC

Signature of Program Head

Signature of Dean